

Video 15 - Kernels

Video Contents

Introduction	06:10-07:31
Kernel as an inner product in some space	07:31-10:06
We only need inner-products	10:06 -12:16
The Kernel Trick	12:16 – 18:44
<u>SAFE SKIP:</u> The Polynomial Kernel	18:44-22:20
Moving to infinite dimensions	22:20-27:55
The Radial Basis Function Kernel	27:55-33:47
<u>SAFE SKIP:</u> Combining the kernel and the SVM algorithm	33:47-39:27
<u>SAFE SKIP:</u> Valid Kernels	39:27-45:44
Soft-margin	45:44-51:45
<u>SAFE-SKIP:</u> Soft-margin maths	51:45-1:01:32
Margin Support Vectors vs Non-Margin Support Vectors	1:01:32- 1:05:15
<u>SAFE SKIP:</u> Two technical remarks	1:05:15-1:09:21
<u>SAFE SKIP:</u> Questions	1:09:21-end

Video Synopsis

Introduction	06:10-07:31
Introduction to the lecture consisting of two topics: Kernels will be discussed which allow the classification problem to be cast in a higher-dimensional space. The soft-margin is introduced as a method of allowing the SVM to misclassify some data points during training to benefit the out-of-sample performance. In practice, you'll likely use both techniques at the same time.	
Kernel as an inner product in some space	07:31-10:06
The Kernel is introduced as a transformation to a Z-space in which we only take the inner-product between different training samples transformed to that Z-space. As the inner product is a scalar, the dimensionality of the Z-space does not come into play (create havoc) and can potentially be infinite. The Kernel trick is introduced as the ability to calculate the inner product of two training data points (vectors) in this Z-space without actually having to transform the two data points to this Z-space.	
We only need inner-products	10:06 -12:16
SAFE SKIP: The mathematics supporting the conclusion that we only need inner-products in the Z-space are introduced.	
The Kernel Trick	12:16 – 18:44
This section introduces the essence of the Kernel trick! It starts by introducing and calculating the inner product of a second order polynomial of an $\mathbf{x}$ and an $\mathbf{x}'$ explicitly. This serves to illustrate what is happening next: a potential kernel expressed in the original space as $(1 + \mathbf{x}^T \mathbf{x}')$ is introduced and it is shown that this expression, which can easily be evaluated (i.e. values calculated for different $\mathbf{x}$ and $\mathbf{x}'$, is actually an inner-product in a higher dimensional space. The conclusion is that we can easily compute the inner product of the vector in the higher dimensional space without ever performing the transformation to this higher dimensional space, so we will never incur the penalty of having to perform the inner product in this higher dimensional space..	
<u>SAFE SKIP:</u> The Polynomial Kernel	18:44-22:20
The equation $(1 + \mathbf{x}^T \mathbf{x}')$ from the previous section, when raised to the power of Q is introduced as the Polynomial kernel. Two hyper parameters for this Kernel are introduced as a and b in the equation: $(a\mathbf{x}^T \mathbf{x}' + b)$ which allow the Kernel to be scaled. In Scikit-learn these are called gamma and coef0 respectively.	
Moving to infinite dimensions	22:20-27:55

It is not necessary to fully understand the mathematics, but it is useful to obtain the intuition that the Kernel introduced in this section, and which can be evaluated very easily in the original space, equates to an inner product of two vectors in an infinitely dimensional space! The consequence of this is that we can leverage the benefits of an infinitely dimensional space (think of linear separability becoming better in higher dimensions) without paying the (infinite) price of having to do calculations in the infinitely dimensional space.

The Radial Basis Function Kernel

27:55-33:47

The Kernel from the last section is introduced as the Radial Basis Function (RBF) kernel and an example application is shown. From the example it is clear, that the RBF kernel is indeed able to come up with a 'wiggly' decision boundary in the original space. The support vectors in this space and their location relative to the decision boundary should not be interpreted in terms of the margin! The margin was determined in the higher dimensional Z-space and is not visible in the original space. What we can see though, is that the number of support vectors is limited and from the previous lecture we know this provides us with an upper estimate of E_{out} ($E(E_{out}) = \frac{E(\#SV)}{N}$). In other words, we are not concerned with the dimensionality of the Z-space, we are only concerned about the number of support vectors found.

SAFE SKIP: Combining the kernel and the SVM algorithm

33:47-39:27

This section shows how the SVM algorithm is adapted using the Kernel and concludes that the only difference is that the Kernel (applied to the training data) replaces the training data itself. This is automatically done for you when you use a Kernel in SVM's.

SAFE SKIP: Valid Kernels

39:27-45:44

Three methods to come up with valid Kernels are introduced: 1) Kernels can be designed by construction as demonstrated for the polynomial Kernel in this video. 2) Using mathematical properties (Mercer's condition). 3) By trial (of a Kernel) and error (see if it works in a classification exercise). The mathematical route via Mercer's condition is further discussed by introducing some mathematical properties that need to be met.

Soft-margin

45:44-51:45

The soft-margin is introduced as dealing with slight issue around non-separability whereas Kernels are used to tackle the severe non-linear separability, for example due to the classification problem being inherently non-linear. When using a soft-margin, you allow certain data points to lie within the margin. To do this, you introduce a 'slack' variable, ξ which measures how far into the margin points lie. The idea of a soft-margin is now to provide a budget for all the points that lie in the margin. In other words, for a soft-margin we will allow the sum of all ξ to be some number greater than 0, but smaller than C.

SAFE-SKIP: Soft-margin maths

51:45-1:01:32

In this section the budget with respect the slack variable ξ is introduced in the Lagrangian to be solved to arrive at the support vectors and it is shown that what appears to be a rather scary addition, results in one minor modification to the original optimisation problem: where previously the Lagrange multipliers α were restricted to being greater than 0, they now also have to be smaller than C, the total budget for the sum of slack variables.

Margin Support Vectors vs Non-Margin Support Vectors

1:01:32- 1:05:15

Margin support vectors are introduced as those support vectors lying exactly on the margin. For these support vectors, the corresponding α will be strictly smaller than C. The other type of support vector is the non-margin support vector. These can lie inside the margin on either side of the margin or even outside the margin but on the opposite side! For these support vectors, the α will always be equal to C.

SAFE SKIP: Two technical remarks

1:05:15-1:09:21

Two technical remarks:

1. The question is raised what results will be obtained from the hard-margin approach for a non-linearly separable data set. The answer is simply that the result cannot be trusted as

a non-linearly separable dataset violates the assumptions made in deriving the hard-margin algorithm.

2. The observation is made that, in transforming data to a higher dimensional features space, constants terms may be introduced that look like the bias term in a features space (i.e. the term that is normally associated with w_0). In the derivation of the SVM algorithm, this term was taken separate into the term 'b', so the question is now, how should we treat these potential constant terms? The answer is that the α associated with any constant term, will go to 0.

SAFE SKIP: Questions

1:09:21-end